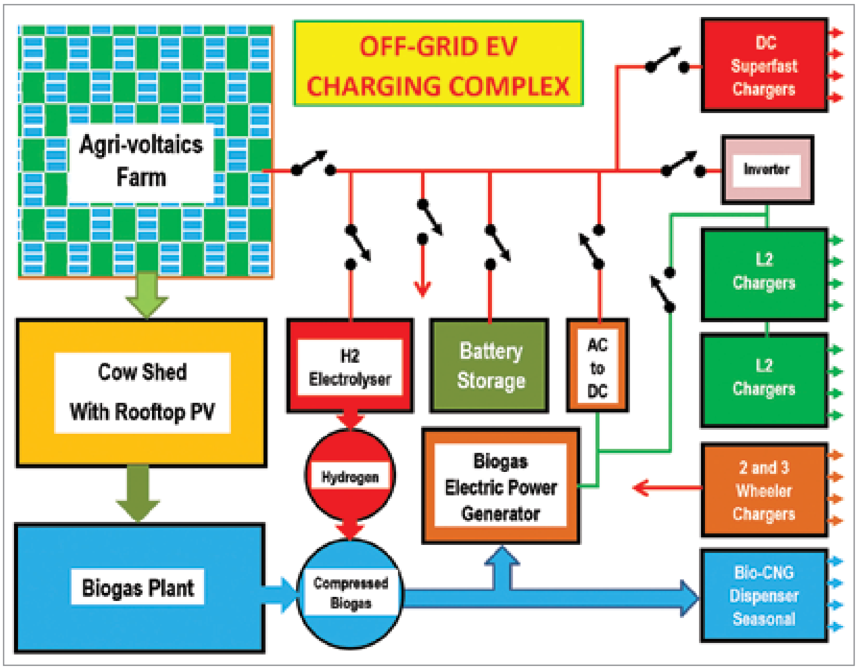

Fig. 3: Layout of proposed EV charging complex

for lithium-ion batteries is increasing, the prices of metals like lithium, cobalt, and nickel are shooting up, resulting in increased price of lithium-ion batteries. To overcome this problem, alternatives such as gravity batteries are being tested.

A heavy block of cement is raised using a crane or elevator mechanism. The block acquires potential energy proportional to the height to which it is raised. When power is needed, the block is slowly lowered, which drives the generator. Such batteries are cost-effective and have life span of more than 25 years.

Here is an example of the gravity battery's capacity:

1. Potential energy = [(mass × acceleration due to gravity × height)/3600]Wh
2. Mass in kg, g = 9.81m/sec^2, height in metres
3. For 100-ton slab, raised to 100m height, energy stored = 27.25kWh

Hydrogen generation systems. Hydrogen is emerging as an alternative to fossil fuels. To make green hydrogen, renewable energy is used. When current is passed through an electrolyser, it splits the water molecules into hydrogen and oxygen.

Hydrogen has many uses. We can mix hydrogen with CNG up to 20%. This mixture of gases can be used in the same equipment as the original CNG generator or vehicle, without any modification to the engine. Also, the safety precautions remain the same as for pure CNG handling. Thus, whenever renewable energy is available in excess quantity, it can be stored as hydrogen gas.

The above calculations assume average values of the parameters and are for general guidance only. While designing an actual system more accurate data has to be considered. For example, type of PV panel, the average sunlight conditions for the selected region, breed of the cows, etc will have large impact on the accuracy of calculations.

A word of caution. Green hydrogen is an emerging field, and the gas is highly explosive in nature. There are safety concerns while handling hydrogen. Government has announced National Hydrogen Mission. We need to set up safe operating procedures and should have trained manpower to handle the gas. Till then, it is not advisable to go for hydrogen based systems. It may take a few years to reach maturity stage.

System design

The layout of proposed EV charging complex is shown in Fig. 3. The agri-voltaics farm is central to this system, around which the whole complex is designed. This farm generates DC power in daytime from the sunlight. The PV power is supplied to DC superfast chargers, L2 chargers, and 2- and 3-wheeler chargers.

Standard L2 chargers need AC input. Therefore, an inverter is used to convert DC supply to AC before connecting to L2 chargers. The DC and L2 chargers need a few tens of kW, hence these will work only when sunlight is strong. The 2- and 3-wheeler chargers need much less power, so these will start working as soon as the sun rises.

EV charging as a service provides good return on investment as compared to selling energy to the grid.

Whenever there is excess PV power, it can be stored in the batteries. If the DC voltage drops due to clouds, power from the battery can be used to support the supply.

When the batteries get fully charged, excess PV power can be used for hydrogen generation. The electrolyser splits water into hydrogen and oxygen. The hydrogen can be stored in a storage tank. The simplest way to use this hydrogen is to mix it with CNG. Up to 20% hydrogen can be mixed without needing any modification to the CNG system.

As mentioned earlier, green hydrogen technology is in initial stages of development. Handling hydrogen, due to its explosive nature, is very challenging. We have to wait till this technology matures before using it. Hence, in the initial implementation of proposed EV charging system, we